YILDIZ TECHNICAL UNIVERSITY
FACULTY OF CIVIL ENGINEERING
DEPARTMENT OF CIVIL ENGINEERING

THE DESIGN OF A HIGHWAY ROUTE

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In 2020-2021 Fall Semester
Prepared in the Division of Transportation

HIGHWAY ENGINEERING COURSE (INS3441)
TERM PROJECT

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ISTANBUL, 2021

INTRODUCTION

Yildiz Technical University, Department of Civil Engineering, 2019-2020 Fall Semester. This year's homework for the Highway Engineering Course (INS3441) is the design of a highway. A transition design was made by working on certain stages. These stages are primarily Determination of the most propriate route (road axis) by making a zero line study between the points A-C. Preparation of the draft plan and the profile. Application of a superelevation with a transition curve on a horizontal curve of the route (Clothoid will be used as the transition curve). Drawing the clothoid detail and the superelevation diagram, highway revolved about the centerline. Drawing the precise plan and the profile. Drawing the cross-sections and calculating their areas.

GIVEN INFORMATIONS

A- Digital contour map (Scale: 1/2.000)

B- Project criteria

1. Project velocity (Vp) = 70 km/h

2. Roadway width (B) = 10 m (2 x 3.5 +2 x 1.5)

3. Maximum grade (gmax) = % 8

4. Right-of-way width = 40 m

5. Minimum curve length (L) = 50 m

6. Maximum superelevation (qmax) = % 8

7. Maximum widening of the horizontal curve (bw) = 2.0 m

8. Minimum vertical curve radii = RCREST = 4,000 m = RSAG = 2,000 m

CALCULATIONS

1- ZERO LINE STUDY

$S_0 = \frac{\Delta H}{L} \times 100$ $\Delta H_{AC} = Hc - Ha = 730 - 710 = 20\ m$ $L = 613.5\ m$

$S_0 = \frac{20}{613.5} \times 100 = 3.26\% < S_{max} = 8\%$ proper! S_0 : slope of zero line

$L_0 = \frac{\Delta H}{S0} \times 100$ $\Delta H = 2$ (the distance between 2 consecutive contour lines)

$L_0 = \frac{2}{3.26} \times 100 = 61.35$ m $L_p = L_0 \times \frac{1}{2000} = 61.35 \times \frac{1}{2000} = 0.0307$ m

2- ROUT STUDY

- **Alignment lengths:**

$|AS_1|$ = 89.41 m $|S_1S_2|$ = 152.02 m $|S_2S_3|$ = 199.45 m $|S_3C|$ = 213.73 m

- **Calculations of curves parameters:**

1st curve: R_1 = 100 m $\Delta_1 = 35.41°$ $t_1 = 100 \times \tan(\frac{35.41}{2}) = 31.92$ m

$D_1 = \frac{2\times\pi\times100}{360} \times 35.41 = 61.81$ m $b_1 = 100 \times (\sec(\frac{35.41}{2}) - 1) = 4.97$ m

2nd curve: R_2 = 100 m $\Delta_2 = 39.60°$ $t_2 = 100 \times \tan(\frac{39.60}{2}) = 36$ m

$D_2 = \frac{2\times\pi\times100}{360} \times 39.60 = 69.12$ m $b_2 = 100 \times (\sec(\frac{39.60}{2}) - 1) = 6.28$ m

3rd curve: R_3 = 100 m $\Delta_3 = 87.67°$ $t_3 = 100 \times \tan(\frac{87.67}{2}) = 96.01$ m

$D_3 = \frac{2\times\pi\times100}{360} \times 87.67 = 153.01$ m $b_3 = 100 \times (\sec(\frac{87.67}{2}) - 1) = 38.63$ m

- **Km values of special points:**

$(A)_{Km} = 0+000^{00}$

$(\Phi_1)_{Km} = A + (AS_1 - t_1) = 0+057^{48}$ $(B_1)_{Km} = \Phi_1 + D_1/2 = 0+088^{39}$

$(T_1)_{Km} = \Phi_1 + D_1 = 0+119^{29}$ $(\Phi_2)_{Km} = T_1 + (S_1S_2 - t_1 - t_2) = 0+203^{37}$

$(B_2)_{Km} = \Phi_2 + D_2/2 = 0+237^{93}$ $(T_2)_{Km} = \Phi_2 + D_2 = 0+272^{50}$

$(\Phi_3)_{Km} = T_2 + (S_2S_3 - t_2 - t_3) = 0+339^{89}$ $(B_3)_{Km} = \Phi_3 + D_3/2 = 0+416^{40}$

$(T_3)_{Km} = \Phi_3 + D_3 = 0+492^{91}$ $(C)_{Km} = T_3 + (S_3C - t_3) = 0+610^{59}$

3- GROUND LEVELS (BLACK ELEVATIONS)

Cross Section NO.	KM	Black elevation		Cross Section NO.	KM	Black elevation
A	0+000^{00}	730.11		15	0+324^{77}	712
1	0+016^{35}	730.41		Φ2	0+339^{89}	710.77
2	0+033^{71}	730		16	0+355^{68}	710.68
Φ1	0+057^{48}	727.57		17	0+369^{83}	712
3	0+069^{76}	726		18	0+381^{07}	714
B1	0+088^{39}	724.68		19	0+386^{71}	716
H1	0+100^{00}	724.34		H4	0+400^{00}	718.19
4	0+111^{22}	724		20	0+411^{21}	720
Ŧ1	0+119^{29}	723.82		B3	0+416^{40}	720.27
5	0+126^{53}	724		21	0+425^{87}	720
6	0+142^{18}	724.46		22	0+436^{54}	718
7	0+158^{82}	724		23	0+447^{18}	714
8	0+179^{76}	723.46		24	0+454^{37}	712
H2	0+200^{00}	723.41		25	0+464^{37}	710
Φ2	0+203^{37}	723.51		26	0+478^{64}	708.78
9	0+221^{23}	723.77		Ŧ3	0+492^{91}	708.62
B2	0+237^{93}	722.45		H5	0+500^{00}	708.89
10	0+243^{21}	722		27	0+510^{62}	710
11	0+254^{25}	721.39		28	0+525^{63}	711.18
12	0+265^{29}	720		29	0+546^{09}	710.7
Ŧ2	0+272^{50}	718.91		30	0+565^{60}	710
13	0+279^{03}	718		31	0+583^{82}	709.25
14	0+291^{76}	716		H6	0+600^{00}	709.47
H3	0+300^{00}	714.76		C	0+610^{59}	710

4- PRECISE AXIS CALCULATION

	X	Y	ΔX	ΔY	Distance
A	959.71	1141.15			
			88.37	13.59	89.41
S_1	1048.08	1154.74			
			109.05	105.92	152.02
S_2	1157.13	1260.66			
			198.83	15.75	199.45
S_3	1355.96	1276.41			
			8.33	213.56	213.72
C	1347.63	1489.97			

$\alpha = \arctan(\frac{\Delta X}{\Delta Y})$; $\alpha_1 = 81.26°$ $\alpha_2 = 45.83°$ $\alpha_3 = 85.47°$ $\alpha_4 = 2.23°$

$\Delta_1 = \alpha_1 - \alpha_2 = 35.43°$ $\Delta_2 = \alpha_3 - \alpha_2 = 39.64°$ $\Delta_2 = \alpha_3 + \alpha_4 = 87.70°$

$|AS_1| = 89.41$ m $|S_1S_2| = 152.02$ m $|S_2S_3| = 199.45$ m $|S_3C| = 213.72$ m

5- CLOTHOID DETAILS

- The clothoid will be applied on the 1st curve:

R_1 = 100 m Δ_1 = 35.41° t_1 = 31.92 m

D_1 = 61.81 m b_1 = 4.97 m

$q_p = 0.443 \times (\frac{V_P^2}{R})$ → $q_p = 0.443 \times (\frac{70^2}{100}) = \%21.71 > \%8$ $q_p > q_{max}$

New project speed: $V_N = \sqrt{\frac{100 \times 8}{0.443}} = 42.50$ Km/h

- Selection of Clothoid parameters:

- Optical condition: $100/3 \leq A \leq 100$

- Dynamic condition: $A_{min} = 0.19 \times \sqrt{V_N^3} = 0.19 \times \sqrt{42.50^3} = 52.64$ m

- Min. Length control: since $L_g < 45$; L_g = 45m will be selected, hence

$A_{min} = \sqrt{L_{gmin} \times R} = \sqrt{45 \times 100} = 67.08$ m ; for safety A_{min} = 70 m

$A_{min}^2 = R \times L_g$ => $70^2 = 100 \times L_g$ => L_g = 49 m proper!

- Geometrical condition: $49/2 \leq 61.81/2$ => $24.5 \leq 30.91$ proper!

- Jerk control: $P' = \frac{V_N^3}{46.7 \times R \times L_g} = \frac{42.50^3}{46.7 \times 100 \times 49} = 0.34 < 0.6$ proper!

Hence Clothoid parameters are: A= 70 m L_g= 49 m

- Calculation of Clothoid members:

$X = L - \frac{L^5}{40 \times A^4} + \frac{L^9}{3456 \times A^8}$ $Y = \frac{L^3}{6 \times A^2} - \frac{L^7}{336 \times A^6} + \frac{L^{11}}{42240 \times A^{10}}$

X and Y values were found at 5 m intervals, from 0 up to L=49.00 m and a clothoid line was formed by connecting these points.

- for L=5m; $X = 5 - \frac{5^5}{40 \times 70^4} + \frac{5^9}{3456 \times 70^8} = 5$

$Y = \frac{5^3}{6 \times 70^2} - \frac{5^7}{336 \times 70^6} + \frac{5^{11}}{42240 \times 70^{10}} = 0$

- for L=49m; $X = 49 - \frac{49^5}{40 \times 70^4} + \frac{49^9}{3456 \times 70^8} = 48.71$

$Y = \frac{49^3}{6 \times 70^2} - \frac{49^7}{336 \times 70^6} + \frac{49^{11}}{42240 \times 70^{10}} = 3.98$

L (m)	X (m)	Y (m)
0	0	0
5	5	0
10	10	0.03
15	15	0.11
20	20	0.27
25	24.99	0.53
30	29.97	0.92
35	34.95	1.46
40	39.89	2.17
45	44.81	3.09
50	49.68	4.23

$\tau = \frac{49}{2\times100}$ = 0.25 rad τ = 14.32°

X_m = 48.71 – 100 x sin(14.32) = 23.98m

Y_m = 3.98 + 100 x cos(14.32) = 100.87m

T_u = 48.71 – 3.98 x cot(14.32) = 33.12m

T_k = 3.98 / sin(14.32) = 16.09m

ΔR = 100.87 – 100 = 0.87m

S = $\sqrt{48.71^2 + 3.98^2}$ = 48.87m

σ = arctan($\frac{3.98}{48.71}$) = 4.67°

$\overline{TS}$ = 23.98 + (100 + 0.87) x $\tan\frac{35.41}{2}$ = 56.18

ΔC = 35.41- 2 x 14.32 = 6.77°

DC = $\frac{2\times\pi\times100\times6.77}{360}$ = 11.82m

- Km values of the beginning and ending points of the Clothoid:

TS_{Km} = $\bar{\Phi}_1$ – L_g/2 = 57.48 – 49/2 = 0+032^{98}

SC_{Km} = $\bar{\Phi}_1$ + L_g/2 = 57.48 + 49/2 = 0+081^{98}

CS_{Km} = $\bar{\Gamma}_1$ - L_g/2 = 119.29 - 49/2 = 0+094^{79}

ST_{Km} = $\bar{\Gamma}_1$ + L_g/2 = 119.29 + 49/2 = 0+143^{79}

6- SUPERELEVATION DETAILS

- Expansion:

R = 100m < 200m , hence expansion is required!

$b_g = \frac{n \times l^2}{2 \times R} + \frac{0.05 \times V_N}{\sqrt{R}} = \frac{2 \times 12^2}{2 \times 100} + \frac{0.05 \times 42.5}{\sqrt{100}} = 1.65\text{m}$

Ramp length: $K = \frac{2 \times h1 \times L}{h2 - h1}$

$h_1 = q_{min}$ x B/2 = 0.02 x 10/2 = 0.10

$h_2 = q_{max}$ x (B+ b_g)/2 = 0.08 x (10+1.65)/2 = 0.466

$K = \frac{2 \times 0.10 \times 49}{0.466 - 0.10} = 26.78\text{m}$

- Km values of the beginning and ending points of the Ramp:

$K_{1Km} = \Phi_1 - L_g/2 - K = 57.48 - 49/2 - 26.78 = 0+006^{20}$

$M_{1Km} = \Phi_1 - L_g/2 - K/2 = 57.48 - 49/2 - 26.78/2 = 0+019^{59}$

$M_{2Km} = \digamma_1 + L_g/2 + K/2 = 119.29 + 49/2 + 26.78/2 = 0+157^{18}$

$K_{2Km} = \digamma_1 + L_g/2 + K = 119.29 + 49/2 + 26.78 = 0+170^{57}$

- Km values of the new cross-sections:

$1_{Km} = TS + L_g/4 = 0+032^{98} + 49/4 = 0+045^{23}$

$2_{Km} = 0+069^{76}$

- Expansion at critical cross section:

$b_g(TS) = 0$ $b_g(\Phi_1) = 1.65/4 = 0.413\text{m}$ $b_g(SC) = 1.65/2 = 0.825\text{m}$

$b_g(1) = (1_{Km} - TS_{Km}) \times \frac{0.825}{49} = 0.206\text{m}$

$b_g(2) = (2_{Km} - TS_{Km}) \times \frac{0.825}{49} = 0.619\text{m}$

- For the outer edge, the superelevation increases linearly from -2% to +8%.

- At the inner edge, the superelevation remains constant at -2% between K1 and TS cross-sections, and decreases linearly to -8% between TS-SC.

- Path widening was applied between TS-SC and it increases linearly with b=0 in TS cross-section and b=bg(max) in SC cross-section.

- In the axis fixed transfer application, the edge heights are calculated by multiplying the half of the sum of the platform width and the expansion amount of the edge to which it belongs, by the transfer.

- The handover and extension between K1-SC is the same (symmetrical) as the handover and extension between CS and K2.

- In the cross-sections between SC and CS, 8% of the rate is constant, and there is a 2% roof slope in all other cross-sections.

Cross section NO.	**KM**	**Distance from the beginning**	**Superelevation %**		**Elevation difference w.r.to the axis elevation**		
			OE	**IE**	**OE**	**AXIS**	**IE**
K	$0+006^{20}$	0	-2	-2	-0.10	0	-0.10
M	$0+019^{59}$	13.39	0	-2	0	0	-0.10
TS	$0+032^{98}$	26.78	2	-2	0.10	0	-0.10
1	$0+045^{23}$	39.03	3.76	-3.76	0.192	0	-0.192
Φ_1	$0+057^{48}$	51.28	5.44	-5.44	0.283	0	-0.283
2	$0+069^{76}$	63.56	7.06	-7.06	0.375	0	-0.375
SC	$0+081^{98}$	75.78	8	-8	0.466	0	-0.466

7- VERTICAL CURVE CALCULATIONS

G = $g_1 - g_2$ = -0.0328 - 0.09 = -0.1228 ➔ sag curve

g_1 = -0.0328 g_2 = 0.09

L = G × R = 0.1228 × 2000 = 245.6 m

t = $\frac{L}{2}$ == 122.8 m

e = G × $\frac{L}{8}$ = 3.77 m

K = $\frac{1}{2R} = \frac{1}{2 \times 2000} = 2.5 \times 10^{-4}$

P_{km} = 0+610^{59}

P_{kot} = 710 m

$T_{1\,km} = P_{km} - t$ = 0+487^{79} m

$T_{1\,kot} = P_{kot} - g_1 \times t$ = 714.03 m

$B_{km} = P_{km}$ = 0+610^{59}

$B_{kot} = P_{kot} + e$ = 713.77 m

$DN_{km} = T_{1\,km} + \frac{L \times g_1}{G}$ = 0+553^{39}

$DN_{kot} = T_{1\,kot} + \frac{L \times g_1^2}{2G}$ = 718.69 + $\frac{79.4 \times 0.0439^2}{2 \times 0.0397}$ = 707.05 m

Cross section NO.	Km	x (m)	$r = g_1.x$	$T_{1kot} - r$	$y = K \times x^2$	$T_{1kot} - r + y$
T_1	$0+487^{79}$	0.000	0.000	714.030	0.000	714.030
T3	$0+492^{91}$	5.120	0.168	713.862	0.007	713.869
H5	$0+500^{00}$	12.210	0.400	713.630	0.037	713.667
24	$0+510^{62}$	22.830	0.749	713.281	0.130	713.411
25	$0+525^{40}$	37.840	1.241	712.789	0.358	713.147
26	$0+546^{09}$	58.300	1.912	712.118	0.850	712.967
DN	$0+553^{39}$	65.600	2.152	711.878	1.076	712.954
27	$0+565^{60}$	77.810	2.552	711.478	1.514	712.991
28	$0+583^{82}$	96.030	3.150	710.880	2.305	713.186
H6	$0+600^{00}$	112.210	3.680	710.350	3.148	713.497
C	$0+610^{59}$	122.800	4.028	710.002	3.770	713.772

8- RED ELEVATION

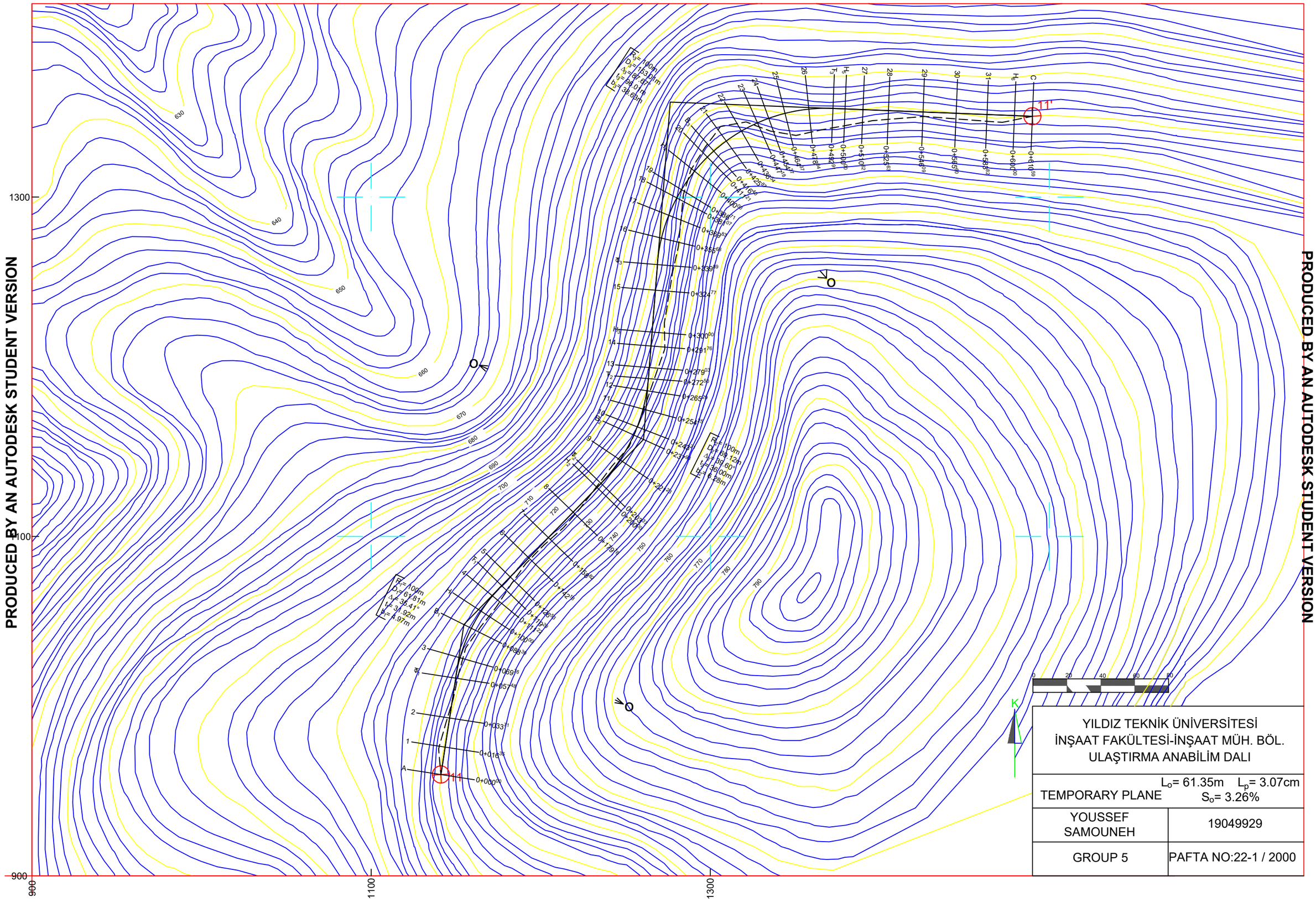
Cross section	Km	Black elevation	Red elevation	Cut	Fill
A	0+.	730.11	730.11	0.00	-
K1	0+6.2	730.45	729.91	0.54	-
M1	0+19.59	730.37	729.47	0.90	-
TS	0+32.98	730.05	729.03	1.02	-
1	0+45.23	728.82	728.63	0.19	-
TB1	0+57.48	727.57	728.23	-	0.66
2	0+69.76	726	727.82	-	1.82
SC	0+81.98	724.99	727.42	-	2.43
B1	0+88.39	724.68	727.21	-	2.53
CS	0+94.79	724.48	727.01	-	2.53
H1	0+100.	724.34	726.83	-	2.49
3	0+111.22	724	726.47	-	2.47
TF1	0+119.29	723.82	726.20	-	2.38
4	0+126.53	724	725.97	-	1.97
ST	0+143.79	724.42	725.40	-	0.98
M2	0+157.18	724	724.96	-	0.96
K2	0+170.57	723.64	724.52	-	0.88
5	0+179.76	723.46	724.22	-	0.76
TB2	0+203.37	723.51	723.45	0.06	-
6	0+221.23	723.77	722.86	0.91	-
B2	0+237.93	722.45	722.32	0.13	-
7	0+243.21	722	722.14	-	0.14
8	0+254.25	721.39	721.78	-	0.39
9	0+265.29	720	721.42	-	1.42
TF2	0+272.5	718.91	721.18	-	2.27
10	0+279.03	718	720.97	-	2.97
11	0+291.76	716	720.55	-	4.55
H3	0+300.	714.76	720.28	-	5.52

12	0+324.77	712	719.47	-	7.47
TB3	0+339.89	710.77	718.98	-	8.21
13	0+355.68	710.68	718.46	-	7.78
14	0+369.83	712	718.00	-	6.00
15	0+381.07	714	717.63	-	3.63
16	0+386.71	716	717.44	-	1.44
H4	0+400.	718.19	717.01	1.18	-
17	0+411.21	720	716.64	3.36	-
B3	0+416.4	720.27	716.47	3.80	-
18	0+425.87	720	716.16	3.84	-
19	0+436.54	718	715.81	2.19	-
20	0+447.18	714	715.46	-	1.46
21	0+454.37	712	715.23	-	3.23
22	0+464.37	710	714.90	-	4.90
23	0+478.64	708.78	714.43	-	5.65
T1	$0+487^{79}$	708.53	714.030	-	5.50
TF3	$0+492^{91}$	708.62	713.869	-	5.25
H5	$0+500^{00}$	708.89	713.667	-	4.78
24	$0+510^{62}$	710	713.411	-	3.41
25	$0+525^{40}$	711.18	713.147	-	1.97
26	$0+546^{09}$	710.7	712.967	-	2.27
DN	$0+553^{39}$	710.41	712.954	-	2.54
27	$0+565^{60}$	710	712.991	-	2.99
28	$0+583^{82}$	709.25	713.186	-	3.94
H6	$0+600^{00}$	709.47	713.497	-	4.03
C	$0+610^{59}$	710	713.772	-	3.77

9- CONCLUSION

The highway connecting points A and C was made according to the rules and specifications outlined in the class. The highway was designed to meet the criteria of comfort for drivers and be economically feasible at the same time. The cut and fill earthworks were almost balanced wich would reduce the cost of construction. Overall, the project helped me get an insight into how highways are designed and constructed, and it made me appreciate the work and the effort of highway engineers in designing safe, comfortable, and economical highways.

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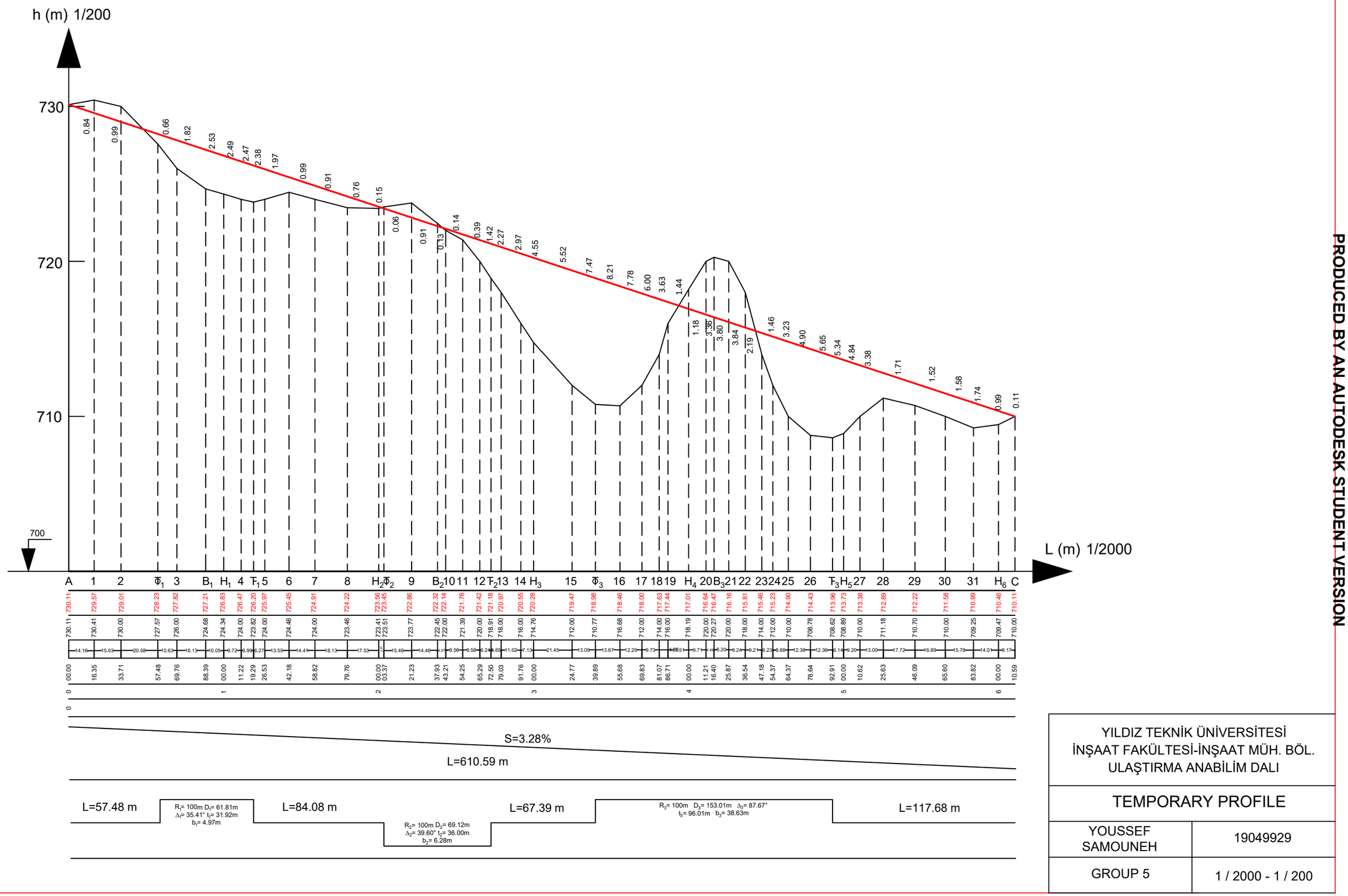


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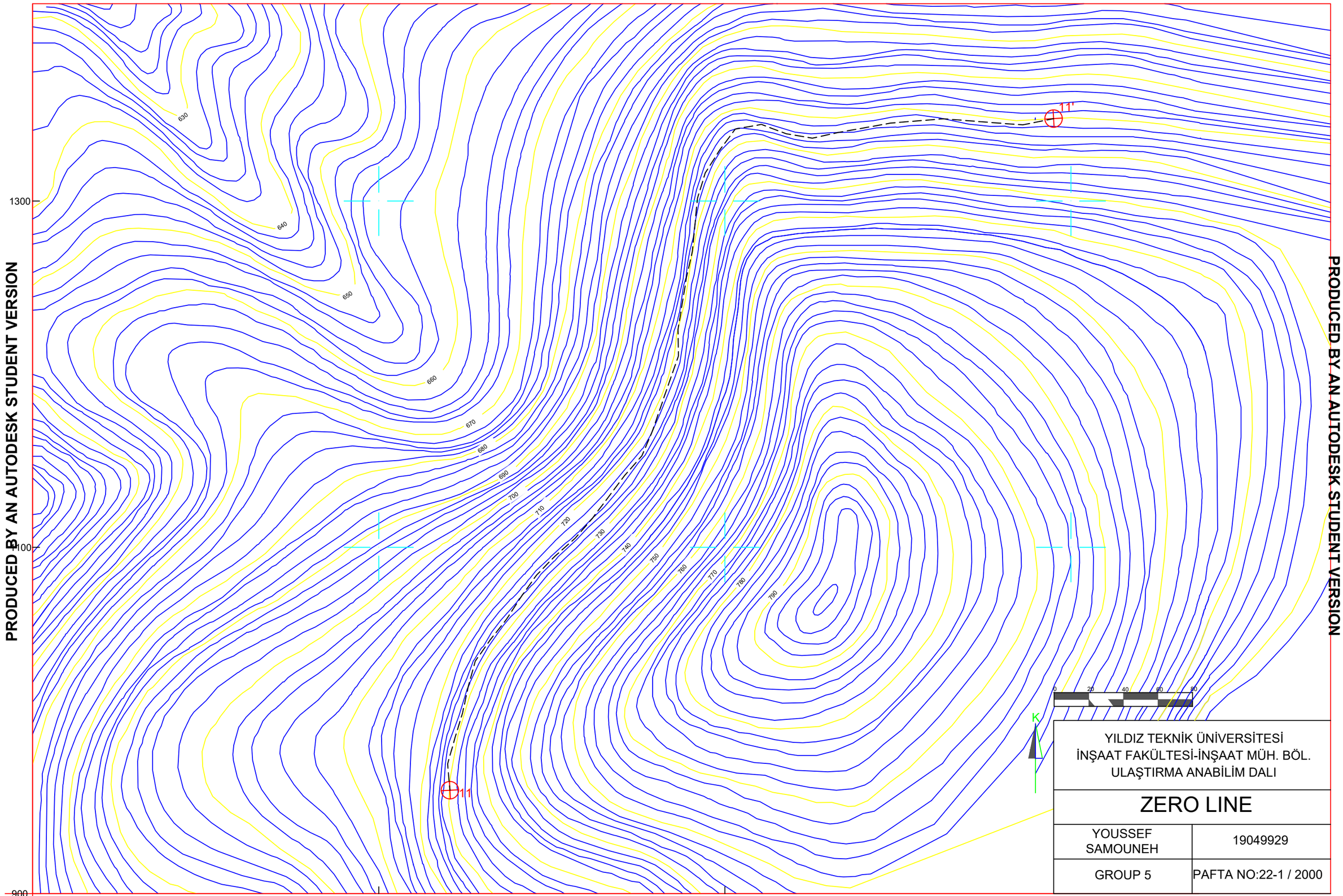
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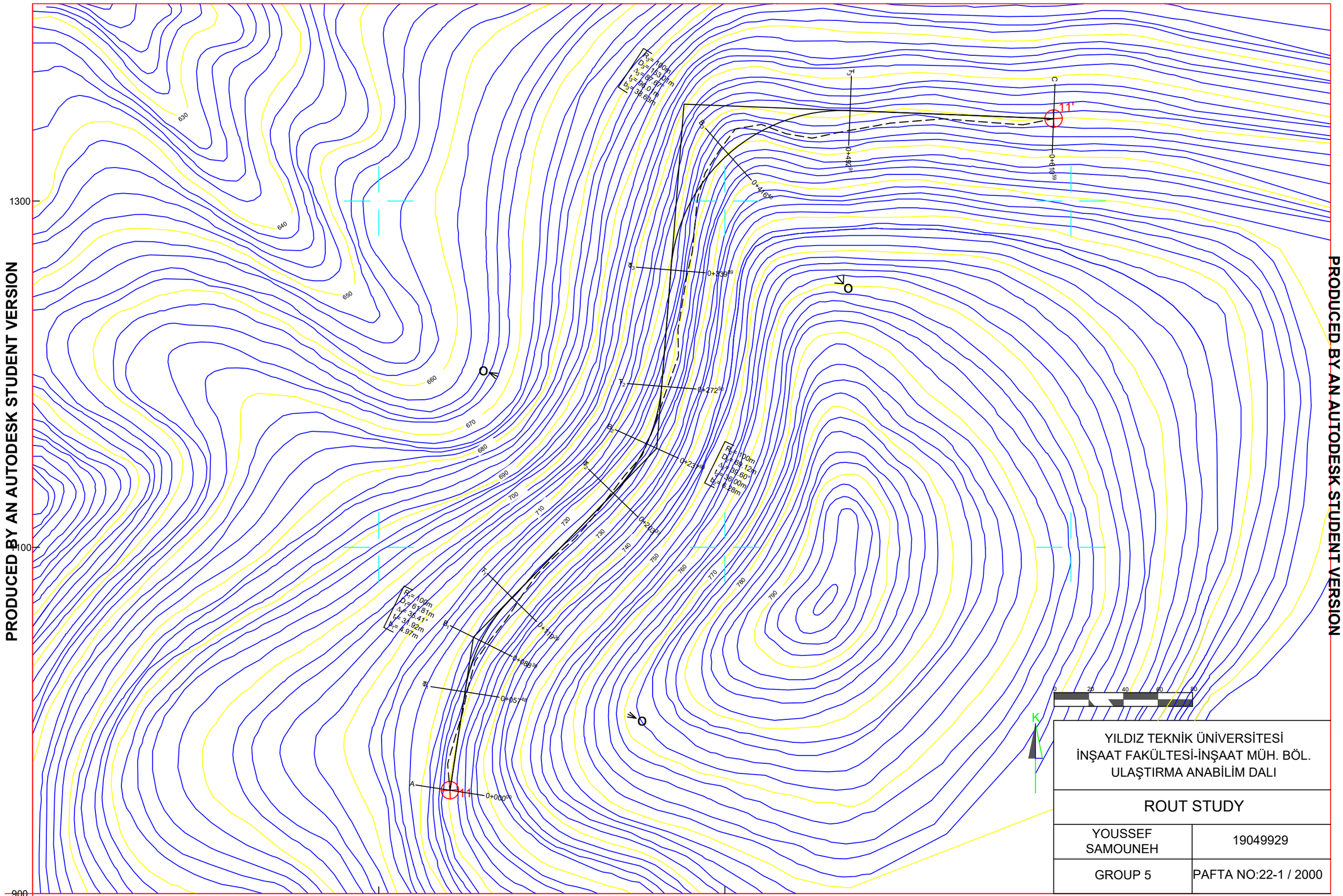
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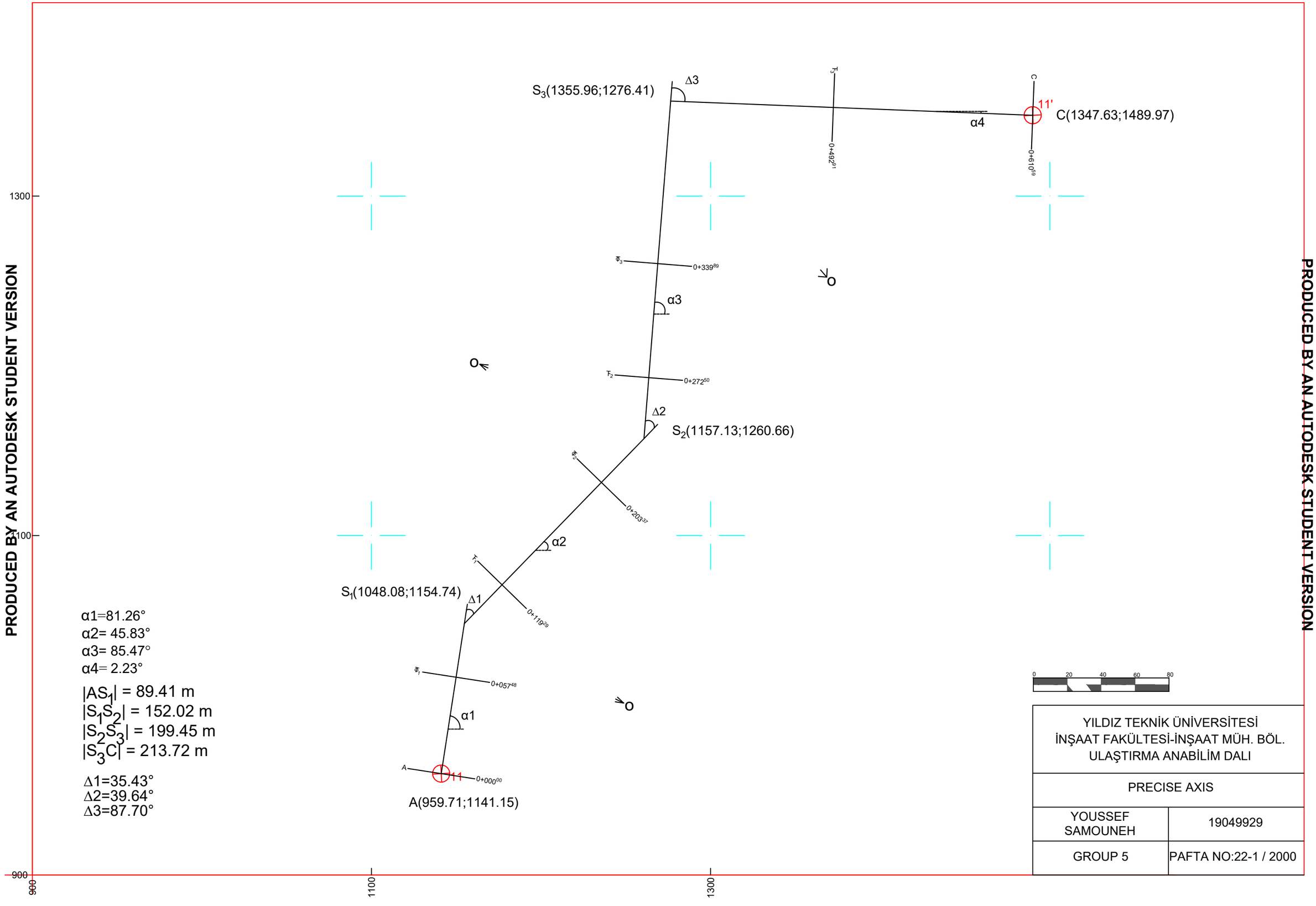
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ZERO LINE	
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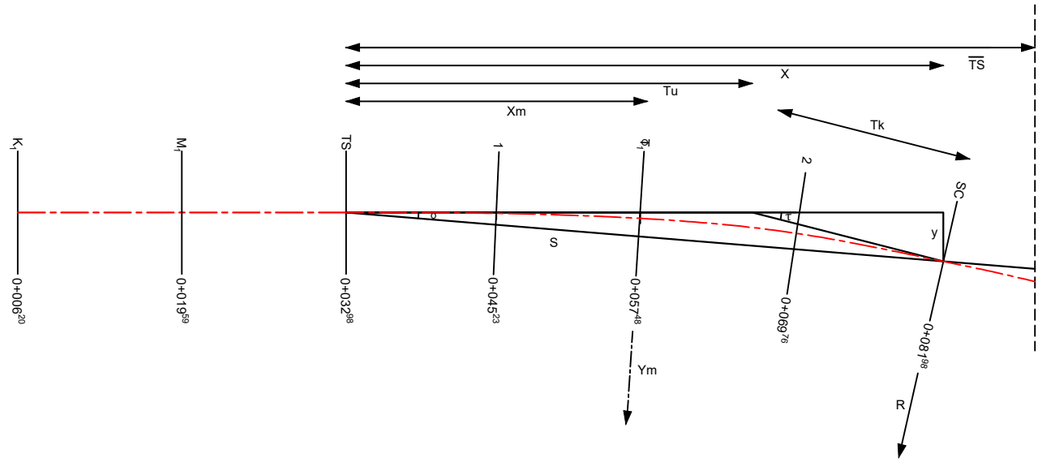


YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
PRECISE AXIS	
YOUSSEF SAMOUNEH	19049929
GROUP 5	PAFTA NO:22-1 / 2000

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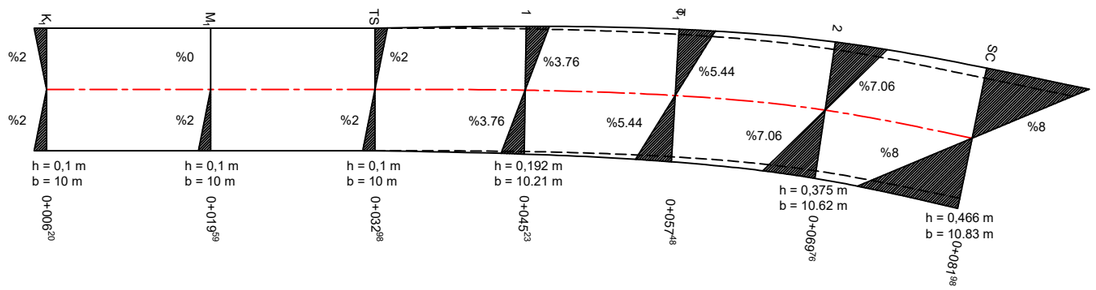


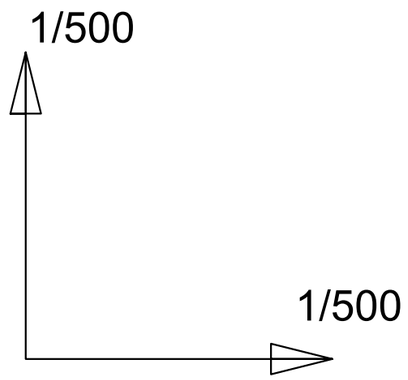
R = 100m
L = 49m
A = 70m
X = 48.71m
Y = 3.98m
$\tau = 14.32°$
$X_m = 23.98m$
$Y_m = 100.87m$
$T_u = 33.12m$
$T_k = 16.09m$
$\Delta R = 0.87m$
S = 48.87m
$\sigma = 4.67°$
$\overline{TS} = 56.18$
$\Delta C = 6.77°$
DC = 11.82m

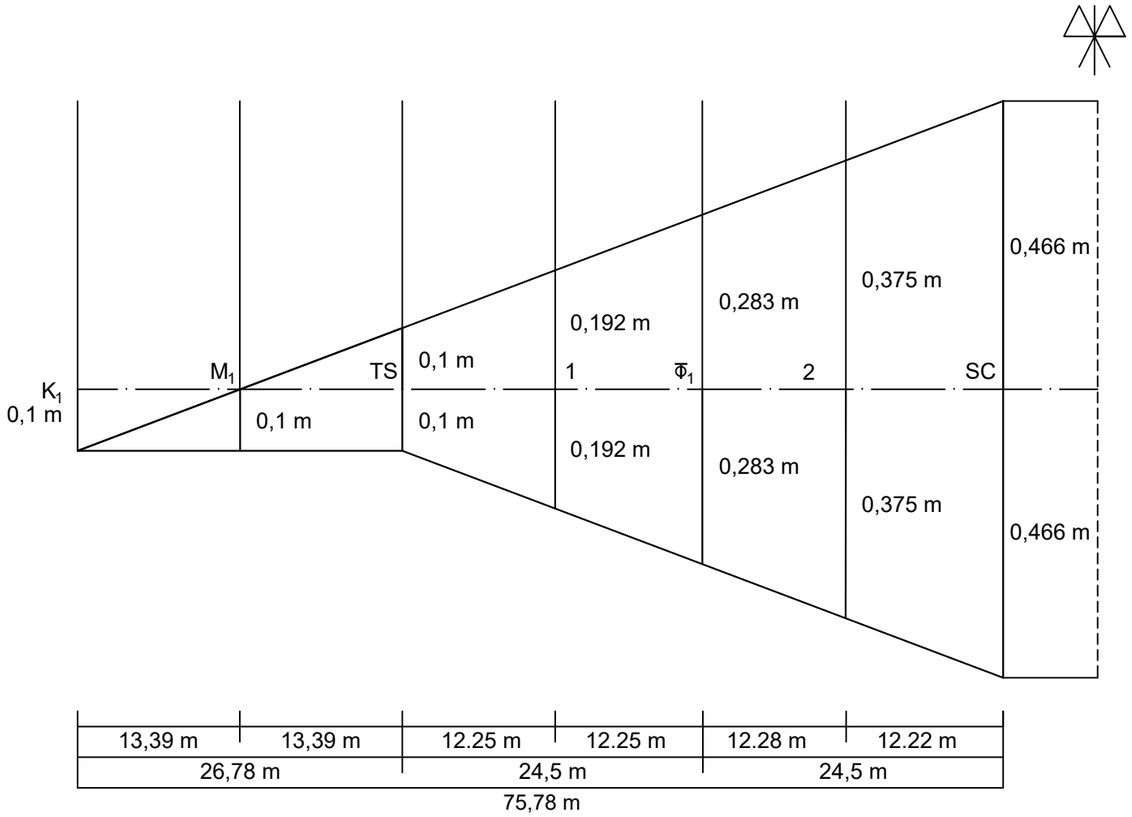
YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
CLOTHOID DETAILS	
YOUSSEF SAMOUNEH	19049929
GROUP 5	SCALE: 1 / 500

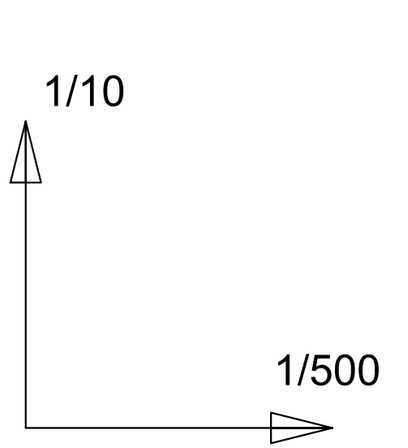
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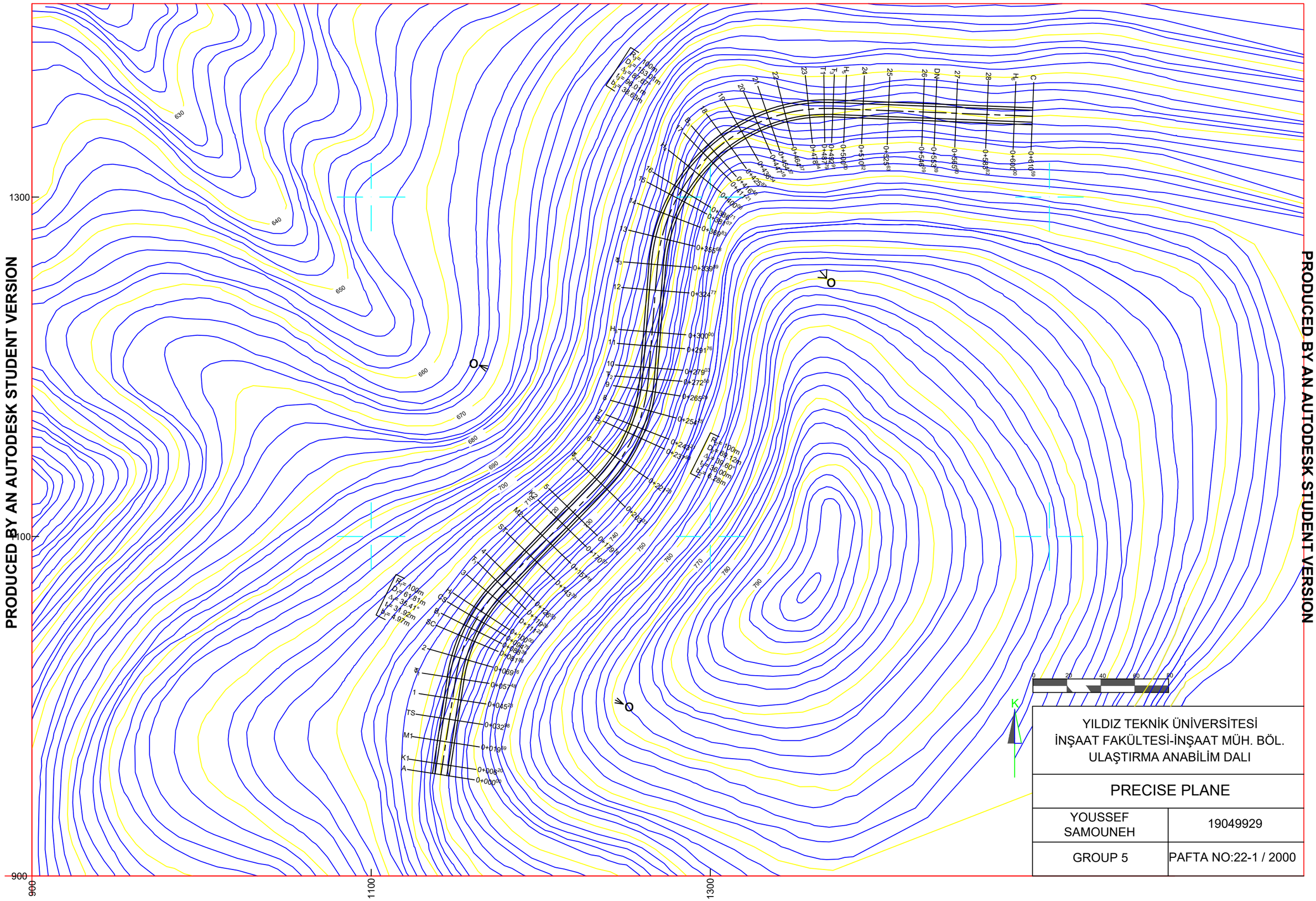


YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
SUPERELEVATION DETAILS	
YOUSSEF SAMOUNEH	19049929
GROUP 5	SCALE: 1 / 500

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YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
PRECISE PROFILE	
YOUSSEF SAMOUNEH	19049929
GROUP 5	SCALE: 1 / 2000 - 1/200

h (m) 1/200

730
720
710
700

L (m) 1/2000

P_{kot} = 710 m
P_{km} = 0+610^{59} m
e = 3,77 m
L = 245,6 m
t = 122,8 m
R = 2000 m

A K1 M1 TS 1 Φ_1 2 SC B_1 C SH_1 3 T_1 4 ST M2 K2 5 Φ_2 6 B_2 7 8 9 T_2 10 11 H_3 12 Φ_3 13 14 15 16 H_4 17 B_3 18 19 20 21 22 23 T_1 T_3 H_5 24 25 26 DN 27 28 H_6 C

730.11 730.45 730.37 730.05 728.82 727.57 726.00 724.99 724.88 724.48 724.34 724.00 723.82 724.00 724.42 724.00 723.64 723.46 723.51 723.77 722.45 722.00 721.39 720.00 718.91 718.00 716.00 714.76 712.00 710.77 710.68 712.00 714.00 716.00 718.19 720.00 720.27 720.00 718.00 714.00 712.00 710.00 708.78 708.53 708.62 708.89 710.00 711.18 710.70 710.41 710.00 709.25 709.47 710.00

6.20 13.39 13.39 12.25 12.25 12.28 12.22 6.41 6.40 5.30 11.22 8.07 7.24 17.26 13.39 13.57 9.19 23.61 17.86 16.70 5.28 11.04 11.04 7.21 6.53 12.73 8.24 24.77 15.12 15.79 14.15 11.24 5.64 13.29 11.21 6.19 9.47 10.67 10.64 7.19 10.00 14.27 9.15 5.12 7.09 10.62 15.01 20.46 7.30 12.21 18.22 16.18 10.59

00.00 06.20 19.59 32.98 45.23 57.48 69.76 81.98 88.39 94.79 00.00 11.22 19.29 26.53 43.79 57.18 70.57 79.76 03.37 21.23 37.93 43.21 54.25 65.29 72.50 79.03 91.76 00.00 24.77 39.89 55.68 69.83 81.07 86.71 00.00 11.21 16.40 25.87 36.54 47.18 54.37 64.37 78.64 87.79 92.91 00.00 10.62 25.63 46.09 53.39 65.60 83.82 00.00 10.59

0 1 2 3 4 5 6

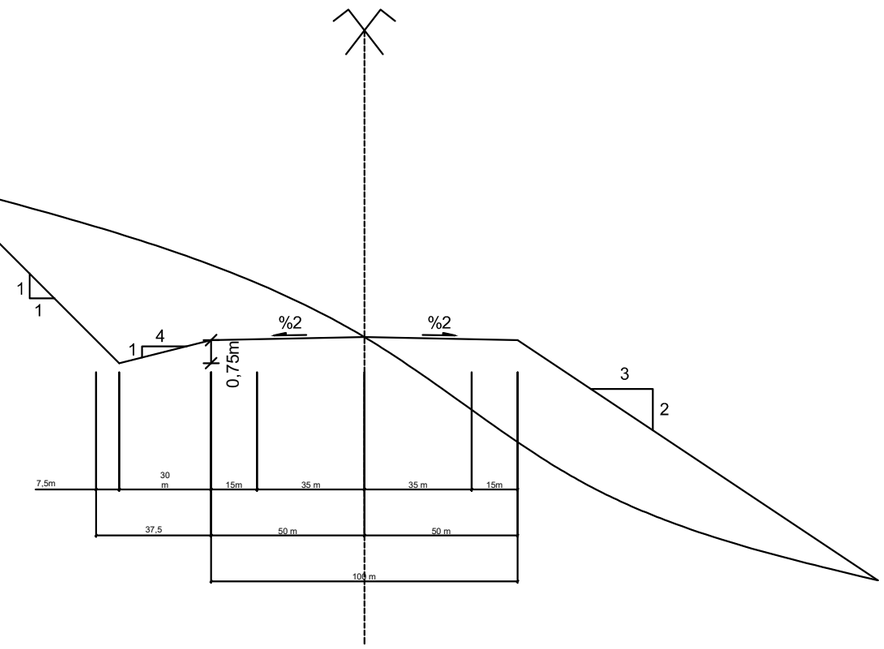
S=3.26%
L = 487.79 m
t = 122.8 m
R = 2000 m

L=32.98 m
R_1= 100m D_1= 61.81m
Δ_1= 35.41° t_1= 31.92m
b_1= 4.97m A=70m Lg=49m
L=59.58 m
R_2= 100m D_2= 69.12m
Δ_2= 39.60° t_2= 36.00m
b_2= 6.28m
L=67.39 m
R_3= 100m D_3= 153.01m Δ_3= 87.67°
t_3= 96.01m b_3= 38.63m
L=117.68 m

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TYPICAL CROSS-SECTION



YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
TYPICAL CROSS-SECTION	
YOUSSEF SAMOUNEH	19049929
GROUP 5	SCALE: 1 / 200

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